

Unsupervised Discovery of Child Welfare Archetypes Across the African Union.

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Abstract. The African Union groups its 55 member states into eight Regional Economic Communities (RECs) for policy coordination, but it is unclear whether these geographic-political blocs reflect the underlying child welfare conditions they are meant to address. Using the 2026 *Africa’s Children Statistical Compendium* (UNICEF), we apply unsupervised pattern recognition over 257 indicators spanning mortality, nutrition, education, WASH, child protection, gender, and economic dimensions. We make two contributions. First, we benchmark six imputation methods on a held-out 15% of known cells and find that low-rank matrix factorization achieves a 17% lower standardized RMSE than a column-mean baseline; the standard iterative (MICE) imputer fails in this small- n /large- p regime ($n = 52$, $p = 257$). Second, we apply PCA followed by k -means, hierarchical (Ward), and Gaussian-mixture clustering, recovering four stable archetypes of child welfare. Bootstrap analysis shows strong reproducibility (within-cluster co-assignment 0.70 vs. 0.13 between). The partition agrees only weakly with REC membership (Adjusted Rand Index 0.10, NMI 0.23), indicating that policy-coordination blocs and child welfare realities are largely orthogonal.

Keywords: Pattern recognition · Unsupervised clustering · Matrix factorization · Imputation · Child welfare · Africa

1 Introduction

The African Union (AU) coordinates its 55 member states largely through eight Regional Economic Communities (RECs) – ECOWAS, SADC, COMESA, EAC, ECCAS, AMU, IGAD, and CEN-SAD – designed around trade, geography, and post-colonial alignments. Substantial child welfare resources are allocated and benchmarked along REC lines [1,2], with the implicit assumption that countries within a REC face sufficiently similar conditions for shared policies to be effective. To our knowledge this assumption has not been tested over the full breadth of child welfare indicators – most existing analyses proceed indicator by indicator or rely on a-priori composite indices. We instead treat AU member states as points in a high-dimensional indicator space and ask: *what natural archetypes of child welfare emerge from the data, and to what extent do these coincide with REC membership?* We contribute (i) a controlled benchmark of six imputation

methods on the compendium’s missingness pattern, including a low-rank matrix factorization tailored to the small- n /large- p regime, and (ii) a clustering analysis using three independent algorithms with bootstrap stability and external validation against REC labels via ARI and NMI.

2 Data and Methods

Data. We use the 2026 *Africa’s Children Statistical Compendium* [1], which aggregates indicators across 21 thematic tables. Source datasets include UNICEF MICS, DHS, WHO modelled estimates, UNESCO UIS, and ILO labour data, with cut-off dates up to April 2026. After parsing multi-row headers and removing regional aggregates, we obtain a country-by-indicator matrix of 52 AU member states and 325 non-empty indicators (overall missingness 25.6%). Restricting to indicators with $\leq 50\%$ missing values yields the working matrix $\mathbf{X} \in \mathbb{R}^{52 \times 257}$ with 14.4% missingness. REC membership was assigned a single primary label per country: ECOWAS (15), COMESA (14), ECCAS (7), EAC (6), AMU (5), SADC (5).

Imputation benchmark. Let Ω be the set of observed cells. We hide a random subset $\Omega_{\text{hold}} \subset \Omega$ of size $0.15|\Omega|$, apply each imputer, and report the standardized RMSE

$$\text{RMSE}_{\text{std}} = \sqrt{\frac{1}{|\Omega_{\text{hold}}|} \sum_{(i,j) \in \Omega_{\text{hold}}} ((\hat{x}_{ij} - x_{ij})/\sigma_j)^2}, \quad (1)$$

where σ_j is the standard deviation of column j on still-known cells. We average over $T = 5$ random splits and compare column mean, column median, k -NN ($k \in \{5, 10\}$), iterative chained-equations imputation [3], and a low-rank matrix factorization (rank $r = 5$) optimised by gradient descent on observed entries with Frobenius regularization [4].

Clustering. After imputation we standardize features and apply PCA, retaining 10 components (73.4% variance). We cluster in this space using k -means (50 restarts), Ward hierarchical clustering, and a Gaussian mixture. We choose $K = 4$ to balance silhouette quality with the constraint $K \geq 4$ for meaningful comparison against the six RECs. Stability is estimated via $B = 50$ bootstrap resamples; we report mean within- and between-cluster co-assignment probability. Agreement with REC labels is measured with Adjusted Rand Index (ARI) [5] and Normalized Mutual Information (NMI).

3 Results

Imputation. Table 1 reports RMSE_{std} across $T = 5$ trials. Matrix factorization achieves 0.913 ± 0.018 , a 16.6% reduction over the column-mean baseline. k -NN underperforms the mean baseline – with only $n=52$ rows, neighbourhoods are

Table 1: Imputation benchmark on 15% held-out cells, 5 random splits.

Method	RMSE _{std}	Std
Column mean	1.101	0.027
Column median	1.142	0.022
k -NN ($k=5$)	1.183	0.034
k -NN ($k=10$)	1.132	0.032
Iterative (MICE)	4.553	0.241
Matrix fact.	0.913	0.018

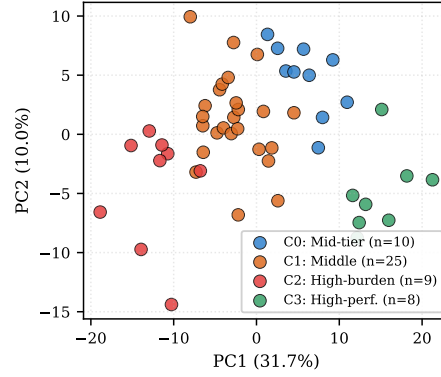


Fig. 1: PCA of the 52 AU states, coloured by data-driven cluster.

noisy. The iterative imputer fails (RMSE 4.55): per-column conditional regressions are ill-posed when $p > n$, and predictions diverge over iterations. This is itself informative: standard iterative methods popular in epidemiology are not directly applicable to compendium-style data without strong feature pre-selection.

Cluster structure. The three clustering algorithms agree well (pairwise ARI 0.42–0.68), indicating the structure is intrinsic rather than algorithm-specific. Bootstrap stability for k -means is high: within-cluster co-assignment 0.704 vs. between-cluster 0.128 (gap 0.576). Figure 1 shows the four archetypes in the PC1–PC2 plane (capturing 41.7% of variance); the same projection coloured by REC (not shown) shows the REC labels disperse across cluster boundaries.

The four archetypes are: **C3 – High-performance** ($n=8$, AMU members plus Cabo Verde, Mauritius, Seychelles – low under-five mortality, high WASH, low fertility); **C0 – Mid-tier** ($n=10$, mostly Southern and East African states); **C1 – Middle** ($n=25$, a large mid-development group spanning West and East Africa); **C2 – High-burden** ($n=9$: Chad, DRC, Eritrea, Ethiopia, Niger, Nigeria, Somalia, South Sudan, Sudan – highest under-five and adolescent mortality, lowest WASH).

Comparison with RECs. The data-driven partition agrees only weakly with REC membership: ARI is 0.104 (k -means), 0.124 (hierarchical), 0.126 (GMM); NMI ranges from 0.227 to 0.267. Inspection of the contingency (Table 2) shows that AMU is the only REC concentrated in a single archetype; all other RECs spread across three or four. ECOWAS distributes its 15 members across all four archetypes, and the high-burden archetype draws members from four different RECs.

Table 2: Cluster–REC contingency.

Cluster	AMU	COMESA	EAC	ECCAS	ECOWAS	SADC
High-burden	0	4	1	2	2	0
High-performance	4	3	0	0	1	0
Mid-tier	0	3	2	2	0	3
Middle	1	4	3	3	12	2

4 Discussion and Conclusion

The principal empirical finding is the disconnect between REC membership and data-driven child welfare archetypes ($\text{ARI} \approx 0.1$). RECs were designed for economic integration, not pediatric outcomes; the magnitude of disagreement, however, has not previously been quantified. Comparative benchmarking along REC lines may therefore obscure the true peer set for any given country: a low-mortality and a high-mortality COMESA country face fundamentally different programming challenges. The high-burden archetype unites countries facing protracted humanitarian crisis (South Sudan, Somalia, Sudan, DRC) regardless of REC affiliation, where coordination on common challenges may be more impactful than within-REC coordination.

Limitations and conclusion. The compendium is a snapshot; longitudinal extension is the natural next step. With $n=52, p=257$ the regime is small- n /high- p , motivating our use of unsupervised methods and stability metrics; supervised modelling here must be approached cautiously. We recovered four stable archetypes of child welfare across the African Union with clear, interpretable indicator profiles. Methodologically, low-rank matrix factorization is well-suited to compendium-style data, materially outperforming column-mean and iterative chained-equations baselines. The archetypes show only weak agreement with the RECs ($\text{ARI} \approx 0.10$), suggesting data-driven peer groupings could complement geographically-defined coordination blocs for child welfare programming.

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